

Academic Council – 18.04.2025

Item No –

As per NEP 2020

Vivek Education Society's

Vivek College of Commerce

(Autonomous)

GOREGAON (WEST)



Syllabus for First Year

Programme: B.Sc. (Information Technology)

With effect from the academic year

2026 – 2027

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Preamble

In today's digital age, information technology has become an integral part of our lives, transforming the way we communicate, learn, work, and interact with the world.

B.Sc. IT degree provides a strong foundation in the theoretical and practical aspects of information technology. The program emphasizes hands-on learning through practical projects, laboratory work, and internships, enabling students to apply their knowledge to real-world scenarios. This practical experience enhances problem-solving skills, fosters creativity, and prepares graduates for the challenges of the IT industry. Upon completion of the B.Sc. IT program, graduates are well-prepared for a wide range of career paths in IT. B.Sc. IT degree empowers graduates with the knowledge, skills, and experience needed to make a meaningful impact in the digital age.

Programme Outcome

PO1 – Possess knowledge to Faculty of Commerce, Science, Arts, and other allied subjects

PO2 – Perform the accomplished tasks with the acquired cognitive and technical skills to evaluate and analyse complex (multifaceted) problems for feasible solutions

PO3 – Apply acquired theoretical, practical, technical, and digital skillset to provide solutions to the problems in the chosen field of discipline

PO4 – Ability to articulate communicate and present their innovative ideas and business models

PO5 – Explore the regulations, universal, human, and ethical values for global well-being

PO6 – Able motivate the research ecosystem to design research proposals and contribute to research in various interdisciplinary domains

PO7 – Coordination and Collaboration with Placement, Career and On-the-job training with external agencies will provide employability to the learners in the chosen field of discipline

Programme Specific Outcome

PSO1: Equip learners with a comprehensive understanding of core IT concepts, including programming languages, software development, database management, networking, web and mobile technologies, artificial intelligence, and other advanced technologies.

PSO2: Cultivate critical thinking and analytical skills to enable students to effectively analyse complex problems, design efficient solutions, and implement them using appropriate technologies.

PSO3: Develop strong communication and interpersonal skills, enabling learners to effectively collaborate with others in diverse teams, articulate technical concepts clearly, and present ideas.

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

PSO4: Promote an understanding of ethical issues related to IT, including privacy, security, and intellectual property. Encourage professional conduct, integrity, and social responsibility in the use of technology.

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

SEMESTER V

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Semester V

Programme Code: - U2IT05			
Course code	Course Type	Course Title	Credits
UIT-MJ351	Major	Advanced ASP.NET with MVC Framework	2
UIT-MJ352	Major	Advanced ASP.NET with MVC Framework Practical	2
UIT-MJ353	Major	Software Project Development	2
UIT-MJ354	Major	Introduction to DevOps Practical	2
UIT-MJ355	Major	Foundation of Indian Logic in Computing	2
UIT-EE351	Elective	Computer Networks	2
UIT-EE352	Elective	Computer Networks Practical	2
UIT-EE353	Elective	Advanced Java Programming	2
UIT-EE354	Elective	Advanced Java Programming Practical	2
UIT-MN351	Minor	MERN Stack Web Development	2
UIT-MN352	Minor	MERN Stack Web Development Practical	2
UIT-VSC351	VSC	Embedded Systems and Internet of Things Practical	2
UIT-	Field Project (FP)		2
		Total	22

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Sr. No.	Course Title	Board of Studies	Page No.
1	Advanced ASP.NET with MVC Framework	Science, Technology and Mathematics	
2	Advanced ASP.NET with MVC Framework Practical	Science, Technology and Mathematics	
3	Software Project Development	Science, Technology and Mathematics	
4	Introduction to DevOps Practical	Science, Technology and Mathematics	
5	Foundation of Indian Logic in Computing	Science, Technology and Mathematics	
6	Computer Networks	Science, Technology and Mathematics	
7	Computer Networks Practical	Science, Technology and Mathematics	
8	Advanced Java Programming	Science, Technology and Mathematics	
9	Advanced Java Programming Practical	Science, Technology and Mathematics	
10	MERN Stack Web Development	Science, Technology and Mathematics	
11	MERN Stack Web Development Practical	Science, Technology and Mathematics	
12	Embedded Systems and Internet of Things Practical	Science, Technology and Mathematics	

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

SEMESTER V

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Advanced ASP.NET with MVC Framework		Course Code: UIT-MJ351	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	--	20

Course Objectives

O1	Understand the basics of .NET Framework and C# programming concepts.
O2	Learn the use of ASP.NET Web Forms and server controls for developing web applications
O3	Understand database connectivity using ADO.NET and data-bound controls.
O4	Introduce advanced ASP.NET concepts such as security, authentication, NuGet, Bootstrap, and AJAX.
O5	Understand the MVC architecture and its working in modern frameworks like Spring MVC.

Module	Course Contents	Lectures	CO Mapping
I	Introduction to .Net Framework and C# Language: .NET Technology and Framework, The C# Language: C# Language Basics, Variables and Data Types, Conditional Logic, Loops, Methods. Types, Objects, and Namespaces: The Basics About Classes, building a Basic Class, Value Types and Reference Types, Understanding Namespaces and Assemblies ASP .Net Controls: ASP.NET Web Forms, Introducing Server Controls, Using the Page Class, Using Application Events, Configuring an ASP.NET Application, AutoPostBack, Using the Validation Controls, Rich Controls: Calendar and AdRotator, Sitemap Path Control. Basic to ADO .Net: ADO.NET Components, Data Providers, SQL Basic command and queries, Understanding the Data Provider Model, Data Controls: Grid View	15	CO1

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

II	Advanced Concepts: NuGet package manager for .NET and Visual Studio- Installation, Basics, framework, Bootstrap basics, AJAX Control Toolkit MVC architecture: Overview of Model-View-Controller, need for separation of concerns, Components of MVC, Working of MVC, Advantages and limitations.	15	CO2, CO3
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Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Beginning Visual C# 2010	K. Watson, C. Nagel, J.H Padderson	Wrox (Wiley)	1 st Edition	2010
2.	Murach's ASP.NET 4.6 Web Programming in C#2015	Mary Delamater and Anne Bohem	SPD	6 th Edition	2016
3.	ASP.NET 4.0 Programming	J. Kanjilal	Tata McGraw Hill	1 st Edition	2011
4.	Programming ML.NET	Dino Esposito, Francesco Esposito	Paperback	1 st Edition	2022
5.	Ajax: A Beginner's Guide	Steven Holzner	Paperback	1 st Edition	2007
6.	Introducing Bootstrap 4	Jörg Krause	Apress	2 nd Edition	2020
7.	Spring MVC Beginner's Guide	Amuthan Ganeshan	Packt Publishing Limited	1 st Edition	2016
Web reference: NuGet package: - https://learn.microsoft.com/en-us/nuget/quickstart/install-and-use-a-package-in-visual-studio					

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Understand fundamentals of .NET Framework and C# programming concepts.	K1, K2	PO1	PSO1
CO2	Apply classes, objects, value/reference types, and namespaces in C#.	K1, K3	PO2, PO3	PSO1, PSO2
CO3	Develop web pages using ASP.NET Web Forms and server controls.	K1, K3	PO2, PO3, PO7	PSO1, PSO2
CO4	Implement database connectivity using ADO.NET and data controls.	K1, K3	PO2, PO3	PSO1, PSO2
CO5	Analyze security, authentication, NuGet, Bootstrap, and AJAX concepts.	K4	PO2, PO5	PSO2, PSO4
CO6	Explain MVC architecture and its components in web applications.	K1, K2	PO1, PO3	PSO1, PSO3

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	0	0	0	0	0	0	3	0	0	0
CO2	0	3	3	0	0	0	0	3	3	0	0
CO3	0	3	3	0	0	0	3	3	3	0	0
CO4	0	3	3	0	0	0	0	3	3	0	0
CO5	0	3	0	0	3	0	0	0	3	0	3
CO6	3	0	3	0	0	0	0	3	0	3	0

Key for Mapping

0 - No correlation 2 - Moderate correlation

1 - Weak correlation 3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Programming with ASP.NET MVC Framework Practical		Course Code: UIT-MJ352	
Periods per week (1 Period is 60 minutes)		4	
Credits		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives

O1	To develop web applications using the .NET framework and C# programming language.
O2	To design interactive web interfaces using ASP.NET server controls, validation controls, and rich controls.
O3	To implement database connectivity and data operations using ADO.NET and SQL queries.
O4	To build and deploy web applications following the MVC architecture with integration of modern tools such as NuGet, Bootstrap, and AJAX.

List of Practical

1.	Write the program for the following:	Lectures	CO Mapping
a.	Create an application to print on screen the output of adding, subtracting, multiplying and dividing two numbers entered by the user in C#.	6	CO1
b.	Create an application to print Floyd's triangle till n rows in C#.		
2.	Write the program for the following:		
a.	Create an application to demonstrate following operations to generate Fibonacci Series.	6	CO2
b.	Create a simple application to demonstrate use of the concepts of interfaces.		
3.	Write the program for the following:		
a.	Create a simple application to demonstrate the concepts boxing and unboxing.	6	CO2
b.	Create a simple application to perform addition and subtraction using delegate.		
4.	Write the program for the following:		
a.	Create a simple web page with various server controls to demonstrate setting and use of their properties.	6	

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

	(Example : AutoPostBack)		
b.	Create a simple application to demonstrate your vacation using calendar control.		CO2
5.	Write the program for the following:		
a.	Create a registration form to demonstrate use of various Validation controls.	6	CO2
b.	Demonstrate the use of Tree view operations on the web form.		
6.	Write the program for the following:		
a.	Create Web Form to demonstrate use of Adrotator Control.	6	CO2
b.	Create Web Form to demonstrate use User Controls		
7.	Write the program for the following:		
a.	Create Web Form to demonstrate use of Website Navigation controls.	6	CO3
b.	Create a web application to demonstrate use of Master Page and content page.		
8.	Write the program for the following:		
a.	Create a web application for inserting and deleting records from a database.	6	CO3
b.	Create a web application to demonstrate use of GridView button column and GridView events along with paging and sorting.		
9.	Write the program for the following:		
a.	Create a web application to demonstrate JS Bootstrap Button.	6	CO3
b.	Create a web application to demonstrate use of various Ajax controls.		
c.	Create a web application to demonstrate Installation and use of NuGet package.		
10.	Write the program for the following:		
a.	Develop a simple Student Registration Web Application using MVC architecture. - Create a Model class to store student details (ID, Name, Email). - Create a View to display a form and show submitted student details. Create a Controller to handle form submission and	6	CO3

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

	send data from the Model to the View.		
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Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Beginning Visual C# 2010	K. Watson, C. Nagel, J.H Padderson	Wrox (Wiley)	1 st Edition	2010
2.	Murach's ASP.NET 4.6 Web Programming in C#2015	Mary Delamater and Anne Bohem	SPD	6 th Edition	2016
3.	ASP.NET 4.0 Programming	J. Kanjilal	Tata McGraw Hill	1 st Edition	2011
4.	Programming ML.NET	Dino Esposito, Francesco Esposito	Paperback	1 st Edition	2022
5.	Ajax : A Beginner's Guide	Steven Holzner	Paperback	1 st Edition	2007
6.	Introducing Bootstrap 4	Jörg Krause	Apress	2 nd Edition	2020
7.	Spring MVC Beginner's Guide	Amuthan Ganeshan	Packt Publishing Limited	1 st Edition	2016
Web reference: NuGet package: - https://learn.microsoft.com/en-us/nuget/quickstart/install-and-use-a-package-in- visual-studio					

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Understand fundamentals of .NET Framework and C# programming concepts.	K1, K2	PO1	PSO1
CO2	Apply classes, objects, value/reference types, and namespaces in C#.	K1, K3	PO2 , PO3	PSO1, PSO2
CO3	Develop web pages using ASP.NET Web Forms and server controls.	K1, K3	PO2, PO3, PO7	PSO1, PSO2
CO4	Implement database connectivity using ADO.NET and data controls.	K1, K3	PO2, PO3	PSO1, PSO2
CO5	Analyze security, authentication, NuGet, Bootstrap, and AJAX concepts.	K4	PO2, PO5	PSO2, PSO4
CO6	Explain MVC architecture and its components in web applications.	K1, K2	PO1, PO3	PSO1, PSO3

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	0	0	0	0	0	0	3	0	0	0
CO2	0	3	3	0	0	0	0	3	3	0	0
CO3	0	3	3	0	0	0	3	3	3	0	0
CO4	0	3	3	0	0	0	0	3	3	0	0
CO5	0	3	0	0	3	0	0	0	3	0	3
CO6	3	0	3	0	0	0	0	3	0	3	0

Key for Mapping

0 - No correlation

2 - Moderate correlation

1 - Weak correlation

3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Software Project Development		Course Code: UIT-MJ353	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	--	20

Course Objectives

O1	To introduce the fundamental concepts and principles of software project planning and management.
O2	To develop understanding of software project estimation, scheduling, and resource planning techniques.
O3	To familiarize students with risk identification, analysis, and mitigation strategies in software projects.
O4	To provide knowledge of monitoring, controlling, and managing software project deliverables throughout the project lifecycle

Module	Course Contents	Lectures	CO Mapping
I	Fundamentals of Software Project Management 1. Introduction to Software Project Management Concept and characteristics of software projects, Importance of software project management, Software project life cycle, Roles and responsibilities of project manager and project team, Stakeholders in software projects 2. Project Initiation and Scope Management Project objectives and scope definition, Project feasibility and business case, Project charter and stakeholder identification, Project success and failure factors 3. Project Planning Project planning activities Select Project, Identify Project Scope and Objectives, S Identify Project Infrastructure, Analyse Project Characteristics, Identify Project Products and Activities, Estimate Effort for Each Activity, Identify Activity Risks, Allocate Resources, Review/Publicize Plan, Execute	15	CO1, CO2

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

	Plan/Lower Levels of Planning, Work Breakdown Structure (WBS), Project scheduling concepts Gantt chart and milestone planning 4. Selection of Software Development Approach Role of development processes in project management Overview of traditional and agile approaches, Factors influencing the selection of development models, Project characteristics affecting development approach 5. Agile Project Management Overview Agile principles, Scrum framework overview, Scrum roles (Product Owner, Scrum Master, Development Team), Scrum artifacts and ceremonies		
II	Estimation, Risk, Tools and Case Studies Software Effort Estimation Estimation challenges, Top-down and bottom-up estimation, Introduction to Function Point Analysis Overview of COCOMO model Project Scheduling Techniques Network planning models, PERT technique Critical Path Method (CPM) Risk Management Risk identification and classification, Risk assessment and mitigation strategies Project Monitoring and Tools Project monitoring and reporting, Earned Value Analysis (concept), Introduction to project management tools- task boards, issue tracking tools, version control platforms Case Studies in Software Project Management Agile project execution example Project failure analysis	15	CO3, CO4

Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Software Project Management	Bob Hughes, Mike Cotterell, Rajib Mall	McGraw Hill Education	6th Edition	2018
2.	Software Project Management in Practice	Pankaj Jalote	Pearson Education	Latest Reprint	2017

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

3.	Software Project Management	Dr. Pallawi Bulakh	Nirali Prakashan	1st Edition	2022
4.	Fundamentals of Software Engineering	Rajib Mall	PHI Learning	5th Edition	2018
5.	Software Engineering: A Practitioner's Approach	Roger S. Pressman, Bruce R. Maxim	McGraw Hill	8th Edition	2015
6.	Project Management: A Systems Approach to Planning, Scheduling, and Controlling	Harold Kerzner	Wiley	12th Edition	2017

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Explain the fundamental concepts, life cycle, and importance of software project management.	K1, K2	PO1, PO2, PO3, PO4, PO5	PSO1, PSO2, PSO3, PSO4
CO2	Apply project initiation and planning techniques such as scope definition, Work Breakdown Structure (WBS), and project scheduling.	K1, K2, K3, K4	PO1, PO2, PO3, PO4	PSO1, PSO2, PSO3
CO3	Analyze software project estimation methods, scheduling techniques (PERT, CPM), and risk management strategies.	K1, K2, K3, K4, K5	PO1, PO2, PO3, PO4, PO6	PSO1, PSO2, PSO3
CO4	Evaluate project performance using monitoring techniques and understand the role of Agile practices and modern project management tools in software projects.	K1, K2, K3, K6	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	2	1	1	0	0	2	2	1	1
CO2	2	3	3	2	0	0	0	2	3	2	0
CO3	2	3	3	1	0	2	0	2	3	1	0
CO4	2	2	3	3	1	1	2	2	2	3	1

Key for Mapping

0- No correlation

2 - Moderate correlation

1- Weak correlation

3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Introduction to DevOps Practical		Course Code: UIT- MJ354	
Periods per week (1 Period is 60 minutes)		4	
Credits		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives

O1	To introduce the fundamental concepts, tools, and practices of DevOps used in modern software development environments.
O2	To develop practical skills in version control systems, continuous integration tools, and automation techniques.
O3	To enable students to build and deploy applications using containerization technologies and DevOps workflows.
O4	To provide hands-on experience in implementing continuous integration and deployment pipelines using modern DevOps tools.

List of Practical:		Lectures	CO Mapping
1.	Introduction to DevOps Tools Objective: Understand DevOps lifecycle and tools.	6	CO1, CO2
a.	Study the DevOps lifecycle.		
b.	Identify common DevOps tools used in industry.		
c.	Install required tools for the lab environment.		
d.	Verify installation of tools.		
2.	Linux Basics for DevOps Objective: Learn basic Linux commands used in DevOps.	6	CO1, CO2
a.	Navigate directories using commands (pwd, ls, cd).		
b.	Create and manage directories and files.		
c.	Display file contents using cat and less.		
d.	Change file permissions using chmod.		

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

3.	Version Control using Git Objective: Learn basic Git operations.	6	CO1, CO2
a.	Configure Git environment.		
b.	Create a local repository.		
c.	Add and commit files.		
d.	View repository history.		
4.	Git Branching and Merging Objective: Work with branches in Git.	6	CO1, CO2
a.	Create branches		
b.	Switch between branches.		
c.	Merge branches.		
d.	Resolve simple merge conflicts.		
5.	Working with Remote Repositories Objective: Connect local repository with remote repository.	6	CO2
a.	Create repository on GitHub.		
b.	Push local code to remote repository.		
c.	Pull updates from remote repository.		
d.	Clone a repository.		
6.	Continuous Integration Setup Objective: Implement basic continuous integration.	6	CO1, CO2
a.	Install Jenkins.		
b.	Configure Jenkins environment.		
c.	Create a basic Jenkins job.		
d.	Execute a build process.		

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

7.	Build Automation Objective: Automate the build process.	6	CO1-CO3
a.	Configure Jenkins with Git repository.		
b.	Trigger builds automatically.		
c.	Monitor build status.		
d.	Analyze build logs.		
8.	Introduction to Docker Objective: Understand containerization.	6	CO1-CO3
a.	Install Docker.		
b.	Run basic Docker commands.		
c.	Pull Docker images.		
d.	Run containers.		
9.	Docker Image Creation Objective: Create and run custom containers.	6	CO1 - CO3
a.	Write a Dockerfile.		
b.	Build a Docker image.		
c.	Run container from created image.		
d.	Manage Docker containers.		
10.	DevOps Pipeline Demonstration & Testing Objective: Implement a simple DevOps workflow.	6	CO1 – CO3
a.	Create a sample application.		
b.	Store code in Git repository.		
c.	Configure Jenkins pipeline.		
d.	Deploy application using Docker container, Deployment on Application Servers		

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	The DevOps Handbook	Gene Kim, Jez Humble, Patrick Debois	IT Revolution Press	2nd	2021
2.	DevOps Adoption Playbook	Sanjeev Sharma	Wiley	1st	2017
3.	Docker Deep Dive	Nigel Poulton	Lean Publishing	Latest	2023
4.	Hands-On DevOps with Linux	Nataraj Das	Packt	1st	2019
References Links: 1. DevOps Overview https://www.atlassian.com/devops 2. Git Documentation https://git-scm.com/docs 3. Docker Documentation https://docs.docker.com 4. Jenkins Documentation https://www.jenkins.io/doc/ 5. Kubernetes Documentation https://kubernetes.io/docs/					

Course Outcomes:

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Understand the DevOps lifecycle and the role of automation in modern software development.	K2	PO1, PO2, PO3, PO4, PO5	PSO1, PSO2

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

CO2	Apply version control practices and collaborative development using Git-based platforms.	K2, K3	PO1, PO2, PO3, PO4	PSO1, PSO2
CO3	Implement continuous integration and build automation using DevOps tools.	K2, K3, K4	PO1, PO2, PO3, PO4	PSO1, PSO2
CO4	Develop and deploy containerized applications using Docker and basic CI/CD pipelines.	K4	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	2	1	1	0	0	3	2	1	0
CO2	2	3	3	2	0	0	0	3	3	2	0
CO3	2	3	3	2	0	0	0	3	3	2	0
CO4	2	2	3	3	1	1	2	3	2	3	1

Key for Mapping

2- No correlation 2 - Moderate correlation

3- Weak correlation 3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Foundation of Indian Logic in Computing		Course Code: UIT- MJ355	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	–	20

Course Objectives: To gain a comprehensive understanding of Indian ideas that can inspire modern information technology.

O1	To introduce students to the basic principles of Ancient Indian logical systems .
O2	To explore logical reasoning, algorithms, and knowledge representation from ancient Indian traditions.
O3	To encourage interdisciplinary thinking between traditional knowledge and contemporary computing technologies .

Module	Course Contents	Lectures	CO Mapping
I	Introduction to Indian Logic: Overview of scientific and technological contributions of ancient India. Role of logic in knowledge and reasoning. Introduction to major schools of Indian philosophy related to logic. Importance of reasoning in ancient texts. Concept of Pramāṇa. Nyāya Logic. Concepts of Hetu (reason) and Vyapti (universal relation). Logical fallacies in Nyāya	15	CO1, CO2
II	Indian Logic and Early Computational Thinking Logical reasoning and decision-making frameworks, Symbolic reasoning and rule-based systems, Logical inference and decision models, Knowledge representation and inference, Simple rule-based systems. Case studies: Future research possibilities of IKS in computing	15	CO2, CO3

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Artificial Intelligence: A Modern Approach	Stuart J. Russell & Peter Norvig	Pearson	4th Edition	2020
2.	The Character of Logic in India	Bimal Krishna Matilal, Jonardon Ganeri & Heeraman Tiwari	SUNY Press	1st Edition	1998
3.	Indian Logic: A Reader	Jonardon Ganeri	Routledge	1st Edition	2001

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Learn about the basic concepts of Indian Knowledge Systems relevant to computing and contributions to logic, mathematics, and algorithms.	K1	PO1, PO2, PO5, PO7	PSO1, PSO2, PSO3
CO2	To relate traditional reasoning systems to modern computing and explore applications of Indian knowledge traditions in modern technological systems.	K2, K3	PO3, PO4, PO6	PSO4, PSO5

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	2	1	1	1	2	1	2	2	1	1	1
CO2	1	2	2	2	1	2	2	1	2	2	2

Key for Mapping

0 - No correlation 2 - Moderate correlation

1 - Weak correlation 3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Computer Networks		Course Code: UIT- EE351	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	--	20

Course Objective:

O1	Understand basic computer networking concepts and devices.
O2	Explain network models and architectures, including OSI and TCP/IP protocol suites.
O3	Analyze the functions of Physical and Data Link Layer.
O4	Understand wireless networking technologies.
O5	Study network layer addressing and routing protocols

Module	Course Contents	Lectures	CO Mapping
I	Introduction: Computer Network, Evolution of Computer Networks, LAN, MAN and WAN, Hardware Devices used for Networking: Network Interface Card (NIC), Modem, Hub, Switch L1 and L2 switches, Comparison between switch and hub, Bridge, Router, Gateway, Topology, Standards and administration. Network Models: Protocol layering, TCP/IP protocol suite, The OSI model.	15	CO1
II	Introduction to Physical layer: Data and signals, periodic analog signals, digital signals, transmission impairment, data rate limits, performance. Introduction to the Data Link Layer: Link layer addressing, Data Link Layer Design Issues, Error detection and correction, block coding Wireless LANs: Introduction, IEEE 802.11 project, Bluetooth, Cellular telephony, Satellite networks. Network Layer: IPv4 Addresses, IPv4 Protocol, ARP, ICMP, IPv6 Routing: RIP, OSPF, BGP Security: Introduction to Network Security and Concepts	15	CO2, CO3, CO4

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Data Communications and Networking	Behrouz A. Forouzan	McGraw-Hill Education	5 th Edition	2013
2.	Computer Networks	Andrew S. Tanenbaum, David J. Wetherall	Pearson	5 th Edition	2011
3.	Network Security Essentials	William Stallings	Pearson	6 th Edition	2020
4.	Computer Networking: A Top-Down Approach	James F. Kurose, Keith W. Ross	Pearson	8 th Edition	2021
Online Resources • https://ekumbh.aicte-india.org/allbook.php • https://free.aicte-india.org/					

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Understand the fundamentals of computer networks, their types (LAN, MAN, WAN), networking devices, topologies, and network models (OSI and TCP/IP).	K1,K2	PO1, PO2	PSO1, PSO2
CO2	Explain the concepts of the physical layer including data and signals, transmission impairments, data rate limits, and performance.	K1,K2	PO1, PO2	PSO1, PSO2
CO3	Describe data link layer functions such as link layer addressing, design issues, and error detection and correction techniques.	K1,K2	PO2, PO3	PSO2, PSO3
CO4	Understand wireless networks, network layer protocols (IPv4, IPv6, ARP, ICMP), and routing protocols like RIP, OSPF, and BGP.	K1,K2,K3	PO2, PO3, PO7	PSO1, PSO2, PSO3

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	0	0	0	0	0	3	2	0	0
CO2	2	2	0	0	0	0	0	2	2	0	0
CO3	0	3	2	0	0	0	0	0	3	2	0
CO4	0	3	3	0	0	0	2	3	3	2	0

Key for Mapping

0 - No correlation

2 - Moderate correlation

1 - Weak correlation

3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Computer Networks Practical		Course Code: UIT- EE352	
Periods per week (1 Period is 60 minutes)		2	
Credits:		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives:

O1	To provide hands-on experience with basic networking concepts, network configuration, and communication protocols.
O2	To develop practical skills in network setup, troubleshooting, and use of networking tools and commands.
O3	To understand network monitoring, packet analysis, and implementation of basic network services through practical experiments

List of Practical:		Lectures	CO Mapping
1.	Study of basic network command and Network configuration commands. ipconfig, netstat, ARP, ping, trace route etc.	6	CO1, CO2
2.	IPv4 Addressing and Subnetting	6	CO1, CO2
a.	Given an IP address and network mask, determine other information about the IP address such as: • Network address • Network broadcast address • Total number of host bits • Number of hosts		
b.	Given an IP address and network mask, determine other information about the IP address such as: • The subnet address of this subnet • The broadcast address of this subnet • The range of host addresses for this subnet • The maximum number of subnets for		

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

	this subnet mask • The number of hosts for each subnet • The number of subnet bits • The number of this subnet		
3.	Designing and configuring a network topology	6	CO1, CO2
a.	Configure IP static routing		
4.	Configure IP routing using RIP.	6	CO1, CO2
5.	Configuring Simple and multi-area OSPF.	6	CO2
6.	Configuring server and client.	6	CO1, CO2
a.	Configure DHCP		
b.	Configure DNS		
c.	Configure HTTP		
d.	Configure Telnet		
e.	Configure FTP		
7.	Configure basic security features for networks.	6	CO1-CO3
8.	Packet capture and header analysis by wire-shark (TCP, UDP, IP etc.)	6	CO1-CO3
Online references link: 1. https://www.cisco.com/site/us/en/learn/topics/small-business/networking-basics.html#tabs-35d568e0ff-item-4bd7dc8124-tab/ 2. https://www.netacad.com/ 3. https://www.wireshark.org/labs/			

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Data Communications and Networking	Behrouz A. Forouzan	McGraw-Hill Education	5 th Edition	2013
2.	Computer Networks	Andrew S. Tanenbaum, David J. Wetherall	Pearson	5 th Edition	2011
3.	Network Security Essentials	William Stallings	Pearson	6 th Edition	2020
4.	Computer Networking: A Top-Down Approach	James F. Kurose, Keith W. Ross	Pearson	8 th Edition	2021

Course Outcomes (CO):

Upon completing this course, the student will be able to:

CO	Course Outcome	Bloom's Taxonomy	PO	PSO
CO1	Use basic network commands for troubleshooting.	K1, K3	PO1, PO2	PSO1, PSO2
CO2	Analyze IPv4 addresses and subnets.	K1, K3	PO2, PO3	PSO1, PSO2
CO3	Design and configure network topologies with RIP/OSPF.	K1, K3	PO2, PO3, PO7	PSO1, PSO2, PSO3
CO4	Configure network services (DHCP, DNS, HTTP, FTP, Telnet).	K1, K4	PO2, PO3	PSO1, PSO2, PSO4
CO5	Capture and analyze network packets.	K1, K3	PO2, PO3, PO6	PSO1, PSO2, PSO4

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	3	2	1	0	0	0	0	3	2	1
CO2	2	3	3	0	0	0	0	0	3	3	0
CO3	2	3	3	2	0	0	3	3	3	3	2
CO4	2	3	3	1	0	0	0	0	3	3	1
CO5	1	3	3	0	0	2	0	2	3	3	1

Key for Mapping

0 – No correlation

1- Weak correlation

2- Moderate correlation

3 – Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B.Sc. (Information Technology)		Semester – V	
Course Name: Advanced Java Programming		Course Code: UIT- EE353	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	--	20

Course Objectives

O1	Understand the concept of Enterprise application development using Java and multi-tier architecture.
O2	Develop dynamic web applications using Java technologies such as Servlets and JSP.
O3	Implement database connectivity in Java applications using JDBC and connection pooling.
O4	Build scalable web applications using frameworks and enterprise technologies.
O5	Understand session management, security, and deployment in enterprise applications.
O6	Design and develop complete enterprise applications using Java web technologies.

Module	Course Contents	Lectures	CO Mapping
I	Introduction: Overview of Enterprise Application Development, Multi-Tier Architecture, Client Server Architecture, Web Servers and Application Servers, HTTP Protocol, Introduction to Java Enterprise Platform. Servlets: Introduction to Servlets, Servlet Architecture, Servlet Life Cycle, Servlet API, Handling Client Requests and Responses, Request Dispatcher, ServletConfig and ServletContext, Session Management (Cookies, URL Rewriting, HttpSession). Java Server Pages (JSP): JSP Architecture, JSP Life Cycle, JSP Directives, JSP Scriptlets, Expressions, JSP Standard Actions, Expression Language (EL), Introduction to JSTL.	15	CO1, CO2

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

II	Database Connectivity: Introduction to JDBC, JDBC Architecture, Types of JDBC Drivers, Establishing Database Connections, Statement and PreparedStatement, Handling ResultSet, Transaction Management, Connection Pooling. Frameworks and Enterprise Development: Dependency Injection Concept, Introduction to Spring Framework, Spring Boot Overview, Creating REST APIs, JSON Data Handling. ORM and Deployment: Introduction to Hibernate ORM, Mapping Objects to Database Tables, CRUD Operations using ORM, Web Application Security Concepts, Authentication and Authorization, Packaging and Deploying Web Applications (WAR files).	15	CO3, CO4
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Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1	Java: The Complete Reference	Herbert Schildt	McGraw-Hill	13 th Edition	2024
2	Core Java Volume II – Advanced Features	Cay S. Horstmann	Pearson	11 th Edition	2020
3	Head First Servlets and JSP	Bryan Basham	O'Reilly	2 nd Edition	2008
4	Spring in Action	Craig Walls	Manning	6 th Edition	2022

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Explain the architecture and components of enterprise web applications including multi-tier architecture, client-server architecture, HTTP protocol, Servlets, and JSP.	K1, K2	PO1, PO2, PO3	PSO1, PSO2, PSO3
CO2	Develop dynamic web applications using Servlets and JSP by handling client requests, responses, and session management	K2, K3	PO1, PO2, PO3, PO4	PSO1, PSO2, PSO3
CO3	Implement database connectivity in enterprise applications using JDBC for executing queries, managing transactions, and handling database results.	K2, K3	PO1, PO2, PO3, PO4	PSO1, PSO2, PSO3
CO4	Build enterprise web applications using frameworks such as Spring and Hibernate, create REST APIs, and deploy secure web applications on application servers.	K4	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4

Mapping- Course outcome (CO) with Programme Specific Outcome (PSO) and Programme Outcome (PO)

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO1	3	2	1	0	0	0	0	3	2	1	0
CO2	2	3	3	2	0	0	0	3	3	2	0
CO3	2	3	3	1	0	0	0	3	3	1	0
CO4	2	2	3	3	1	1	2	3	2	3	1

Key for Mapping

0 – No correlation 1- Weak correlation
2- Moderate correlation 3 – Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Advanced Java Programming Practical		Course Code: UIT- EE354	
Periods per week (1 Period is 60 minutes)		4	
Credits		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives

O1	Understand the concept of Enterprise application development using Java and multi-tier architecture.	
O2	Develop dynamic web applications using Java technologies such as Servlets and JSP.	
O3	Implement database connectivity in Java applications using JDBC and connection pooling.	
O4	Build scalable web applications using frameworks and enterprise technologies.	
O5	Understand session management, security, and deployment in enterprise applications.	
O6	Design and develop complete enterprise applications using Java web technologies.	

List of Practical:		Lectures	CO Mapping
1.	Install and configure Java Enterprise development environment using JDK, IDE and Application Server such as Apache Tomcat. Create and deploy a simple web application.	6	CO1
2.	Develop a basic Servlet program to handle HTTP request and response and display dynamic output on the browser.	6	CO1, CO2
3.	Write a Servlet program to demonstrate Servlet Life Cycle methods (init(), service(), destroy()).	6	CO2
4.	Create a web application using Servlets to demonstrate Session Management using Cookies and HttpSession.	6	CO2, CO5
5.	Develop a JSP page to demonstrate JSP directives, scriptlets, expressions and implicit objects	6	CO2
6.	Design a JSP based form processing application that accepts user input and displays the result dynamically.	6	CO2
7.	Implement JDBC connectivity to connect a Java web application with a database and perform CRUD operations.	6	CO3
8.	Develop a web application using JSP and JavaBeans to separate presentation and business logic.	6	CO3
9.	Create a simple RESTful web service using Spring Boot to perform CRUD operations on database records.	6	CO4, CO6

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

10.	Develop a mini–Enterprise Web Application integrating Servlet/JSP, JDBC and MVC architecture and deploy it on Apache Tomcat.	6	CO4, CO5, CO6
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Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	The DevOps Handbook	Gene Kim, Jez Humble, Patrick Debois	IT Revolution Press	2nd	2021
2.	DevOps Adoption Playbook	Sanjeev Sharma	Wiley	1st	2017
3.	Docker Deep Dive	Nigel Poulton	Lean Publishing	Latest	2023
4.	Hands-On DevOps with Linux	Nataraj Das	Packt	1st	2019

References Links:

6. DevOps Overview
<https://www.atlassian.com/devops>
7. Git Documentation
<https://git-scm.com/docs>
8. Docker Documentation
<https://docs.docker.com>
9. Jenkins Documentation
<https://www.jenkins.io/doc/>
10. Kubernetes Documentation
<https://kubernetes.io/docs/>

Course Outcomes:

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Set up and configure enterprise Java development environment and deploy basic web applications.	K2	PO1, PO2, PO3, PO4, PO5	PSO1, PSO2

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

CO2	Develop dynamic web applications using Servlets and JSP for handling HTTP requests, responses and session management.	K3	PO1, PO2, PO3, PO4	PSO1, PSO2
CO3	Implement database connectivity using JDBC and perform CRUD operations in enterprise applications.	K3	PO1, PO2, PO3, PO4	PSO1, PSO2
CO4	Develop enterprise applications using modern frameworks such as Spring Boot and create RESTful services.	K4	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4
CO5	Integrate multiple web technologies to build complete enterprise web applications.	K5	PO1, PO2, PO3, PO4, PO6	PSO1, PSO2, PSO3
CO6	Deploy and manage enterprise applications on application servers.	K3, K4, K5	PO1, PO2, PO3, PO4, PO5, PO7	PSO1, PSO2, PSO3

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	1	0	0	0	0	3	2	1	0
CO2	2	3	3	2	0	0	0	3	3	2	0
CO3	2	3	3	1	0	0	0	3	3	1	0
CO4	2	2	3	3	1	1	0	3	2	3	1
CO5	2	2	3	3	1	1	1	3	3	3	1
CO6	2	2	3	2	1	1	2	3	0	0	0

Key for Mapping

0- No correlation

2 - Moderate correlation

1- Weak correlation

3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B.Sc. (Information Technology)		Semester – V	
Course Name: MERN Stack Web Development		Course Code: UIT- MN351	
Periods per week (1 Period is 60 minutes)		2	
Credits		2	
		Hours	Marks
Evaluation System	Theory Examination	1	30
	Internal	--	20

Course Objectives

O1	To introduce the architecture and workflow of MERN-based full stack web applications.
O2	To develop secure and scalable full stack applications using authentication, authorization, and state management techniques.
O3	To implement advanced full stack features such as testing, logging, performance optimization, and deployment strategies.
O4	To build and deploy a complete MERN stack application integrating frontend, backend, database, and cloud services.

Module	Course Contents	Lectures	CO Mapping
I	Full Stack Architecture & Integration Overview of Full Stack Architecture in MERN Connecting React Frontend with Node.js/Express Backend (REST API consumption using Axios/Fetch) Cross-side Scripting essentials JWT Authentication & Authorization Role-Based Access Control (RBAC) Environment variables & configuration management State management with backend data (React Context/Redux basics) File uploads using Express & cloud storage basics	15	CO1, CO2

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

II	Advanced Full Stack Development & Deployment	15	CO3, CO4
	Web application security best practices (XSS, CSRF basics) Performance optimization in MERN applications Logging & monitoring basics Testing (Unit testing and API testing basics) CI/CD basics for MERN applications Docker basics for dynamic application containerization Deployment to Cloud platforms (Render / Vercel / similar platforms) Capstone MERN Full Stack Project		

Books and References:

Sr. No.	Title	Author/s	Publisher	Edition	Year
1	Pro MERN Stack: Full Stack Web App Development with MongoDB, Express, React, and Node	Vasan Subramanian	Apress	2nd Edition	2019
2	Full-Stack React Projects	Shama Hoque	Packt Publishing	1st Edition	2018
3	Learning React	Alex Banks, Eve Porcello	O'Reilly Media	2nd Edition	2020
4	Node.js Design Patterns	Mario Casciaro, Luciano Mammino	Packt Publishing	3rd Edition	2020
5	MongoDB: The Definitive Guide	Shannon Bradshaw, Kristina Chodorow, Michael Dirolf	O'Reilly Media	3rd Edition	2019
6	Hands-On Full Stack Development with MongoDB	Shubham Satyarthi	Packt Publishing	1st Edition	2019

Learning Resources

- React Documentation: <https://react.dev>
- Node.js Documentation: <https://nodejs.org/en/docs>
- MongoDB Documentation: <https://www.mongodb.com/docs>
- Express.js Documentation: <https://expressjs.com>

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Course Outcomes (CO)

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Explain the architecture and components of MERN-based full stack applications.	K1, K2	PO1, PO2, PO3	PSO1, PSO2, PSO3
CO2	Develop secure MERN applications using APIs, authentication mechanisms, and state management.	K3	PO1, PO2, PO3, PO4	PSO1, PSO2, PSO3
CO3	Analyze and implement advanced features such as testing, monitoring, performance optimization, and containerization.	K4	PO1, PO2, PO3, PO4, PO5, PO6	PSO1, PSO2, PSO3, PSO4
CO4	Design and deploy complete full stack applications using MERN technologies and cloud deployment platforms.	K5	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4

Mapping- Course outcome (CO) with Programme Specific Outcome (PSO) and Programme Outcome (PO)

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	1	0	0	0	0	3	2	1	0
CO2	2	3	3	2	0	0	0	3	3	2	0
CO3	2	3	3	2	1	1	0	3	3	2	1
CO4	2	3	3	3	1	1	2	3	3	3	1

Key for Mapping

0 – No correlation 1- Weak correlation

2- Moderate correlation 3 – Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: MERN Stack Web Development Practical		Course Code: UIT- MN352	
Periods per week (1 Period is 60 minutes)		4	
Credits		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives

O1	Understand the architecture and setup of MERN stack applications including frontend, backend, and database integration.
O2	Develop full stack web applications using React, Node.js, Express, and MongoDB with CRUD functionality.
O3	Implement secure web applications using authentication, authorization, and validation techniques.
O4	Integrate, test, and deploy complete MERN stack applications on cloud platforms.

List of Practical:		Lectures	CO Mapping
1	Set up a MERN stack development environment by creating a React frontend application and Node.js + Express backend server.	6	CO1
2	Develop a simple REST API using Express.js and connect the React frontend to fetch and display data using Axios or Fetch API.	6	CO1
3	Create a basic MERN CRUD application implementing Create and Read operations using MongoDB and Mongoose.	6	CO2
4	Extend the MERN CRUD application to support Update and Delete operations with proper API endpoints.	6	CO2
5	Implement user registration and login functionality using Express.js backend APIs and MongoDB database.	6	CO3
6	Secure MERN API endpoints using JSON Web Token (JWT) authentication and authorization mechanisms.	6	CO3
7	Implement form validation and error handling in React frontend and Express backend.	6	CO3
8	Display, filter, and manage database records dynamically using React components and backend APIs.	6	CO4
9	Test and debug MERN APIs using Postman and browser developer tools.	6	CO4

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

10	Build and deploy a complete MERN stack application to a cloud hosting platform such as Render or Vercel.	6	CO4
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Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1	Pro MERN Stack: Full Stack Web App Development with MongoDB, Express, React, and Node	Vasan Subramanian	Apress	2nd Edition	2019
2	Full-Stack React Projects	Shama Hoque	Packt Publishing	1st Edition	2018
3	Learning React	Alex Banks, Eve Porcello	O'Reilly Media	2nd Edition	2020
4	Node.js Design Patterns	Mario Casciaro, Luciano Mammino	Packt Publishing	3rd Edition	2020
5	MongoDB: The Definitive Guide	Shannon Bradshaw, Kristina Chodorow, Michael Dirolf	O'Reilly Media	3rd Edition	2019
6	Hands-On Full Stack Development with MongoDB	Shubham Satyarthi	Packt Publishing	1st Edition	2019
Learning Resources - React Documentation: https://react.dev - Node.js Documentation: https://nodejs.org/en/docs - MongoDB Documentation: https://www.mongodb.com/docs - Express.js Documentation: https://expressjs.com					

Course Outcomes:

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Explain and configure MERN stack development environments and architecture.	K2	PO1, PO2, PO3,	PSO1, PSO2, PSO3

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

CO2	Develop full stack applications using React frontend and Node.js/Express backend with database operations.	K3	PO1, PO2, PO3, PO4	PSO1, PSO2, PSO3
CO3	Implement authentication, authorization, and validation mechanisms in MERN applications.	K4	PO1, PO2, PO3, PO4, PO5	PSO1, PSO2, PSO3, PSO4
CO4	Integrate, test, and deploy full stack applications using cloud platforms and modern development tools.	K5	PO1, PO2, PO3, PO4, PO5, PO6, PO7	PSO1, PSO2, PSO3, PSO4

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	2	1	0	0	0	0	3	2	1	0
CO2	2	3	3	2	0	0	0	3	3	2	0
CO3	2	3	3	2	1	0	0	3	3	2	1
CO4	2	3	3	3	1	1	2	3	3	3	1

Key for Mapping

0-No correlation

1-Weak correlation

2 - Moderate correlation

3 - Strong correlation

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Programme Name: B. Sc. (Information Technology)		Semester – V	
Course Name: Embedded System and Internet of Things Practical		Course Code: UIT- VSC351	
Periods per week (1 Period is 60 minutes)		4	
Credits		2	
		Hours	Marks
Evaluation System	Practical Examination	1	30
	Internal	1	20

Course Objectives

O1	Introduce the fundamental concepts of embedded systems.
O2	Familiarize students with sensors and actuators.
O3	Develop basic programming skills for microcontrollers.
O4	Enable students to interface hardware components.
O5	Introduce IoT architecture and cloud connectivity.
O6	Develop practical skills through implementation of a mini IoT project.

List of Practical:		Lectures	CO Mapping
1.	Fundamentals of Embedded Systems	6	CO1, CO2, CO3
a.	Installation of Arduino IDE		
b.	Writing and executing first program (LED Blinking)		
c.	Understanding digital input/output		
2.	Sensors and Actuators	6	CO1, CO2, CO3
a.	Interfacing temperature sensor		
b.	Distance measurement using ultrasonic sensor		
c.	Automatic lighting system using LDR		
d.	Controlling buzzer		
e.	Servo motor angle control		
f.	DC motor control using motor driver		

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

3.	Microcontroller Programming and Interfacing	6	CO1, CO2 CO3
a.	LED pattern generation		
b.	Push button interfacing		
c.	Traffic light simulation		
d.	Temperature-based fan control		
e.	IR-based object counter		
4.	Internet of Things (IoT) Concepts and Connectivity	12	CO1, CO2, CO3, CO4
a.	Interfacing Wi-Fi module (ESP8266/ESP32)		
b.	Sending sensor data to cloud		
c.	Visualizing data on dashboard		
d.	Controlling LED using mobile application		
5.	IoT Applications and Mini Project (Compulsory Internal Activity)	30	CO1, CO2, CO3, CO4, CO5
	<u>Students shall design and implement a basic IoT system incorporating:</u> • At least one sensor • At least one actuator • Internet connectivity • Dashboard visualization Project demonstration and viva-voce are mandatory.		

Books and References:					
Sr. No.	Title	Author/s	Publisher	Edition	Year
1.	Embedded Devices and Internet of Things: Technologies and Applications	Adesh Kumar, Surajit Mondal, Gaurav Verma, Prashant Mani	CRC Press	1st	2024
2.	Internet of Things	Raj Kamal	Tata McGraw Hill Education	2nd	2022
3.	Internet of Things with Arduino	Marco Schwartz	Packt Publishing	1st	2016

FORMAT NO.	REV. NO.	DATE	DEPARTMENT
VES/VCC/ACAD/SYLL/F011	0	08.08.2024	ACADEMIC

Course Outcomes:

Upon completing this course, the student will be able to:

CO	Course Outcomes	Blooms Taxonomy	PO	PSO
CO1	Explain the basic architecture and components of embedded systems.	K2, K3	PO1, PO2, PO3	PSO1, PSO2
CO2	Interface and operate common sensors and actuators	K2, K3	PO1, PO2, PO3	PSO1, PSO2
CO3	Develop simple embedded programs using Arduino IDE.	K1, K2, K6	PO1, PO2, PO3	PSO1, PSO2
CO4	Transmit sensor data to IoT cloud platforms.	K3, K4, K6	PO1, PO2, PO3	PSO1, PSO2, PSO4
CO5	Design and implement a basic IoT-based application.	K6	PO1, PO2, PO3, PO4, PO5, PO7	PSO1, PSO2, PSO3, PSO4

Mapping- Course outcome with Programme Specific Outcome and Programme Outcome

PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2	PSO3	PSO4
CO											
CO1	3	3	2	0	0	0	0	3	2	0	0
CO2	3	3	2	0	0	0	0	3	2	0	0
CO3	3	2	2	0	0	0	0	3	2	0	0
CO4	3	3	2	0	0	0	0	3	3	0	0
CO5	3	3	3	3	3	0	3	3	3	3	3

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